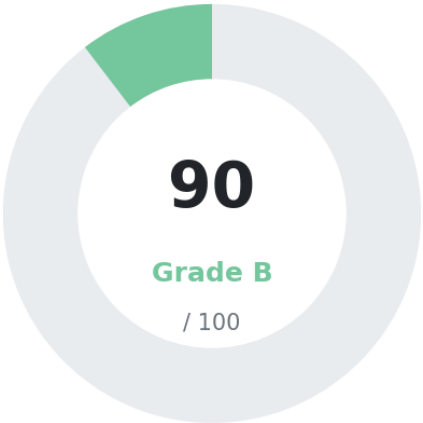
■ dataDoctor

Data Quality Intelligence Report

Dataset: **dataDoctor_big_dataset.csv**

Generated: April 26, 2026 at 16:30

Overall Score



Overall Quality Score: 89.6/100
Good ■ | ■ Good

Shape: 10,300 rows × 8 columns | 4 numeric features

1. Executive Summary

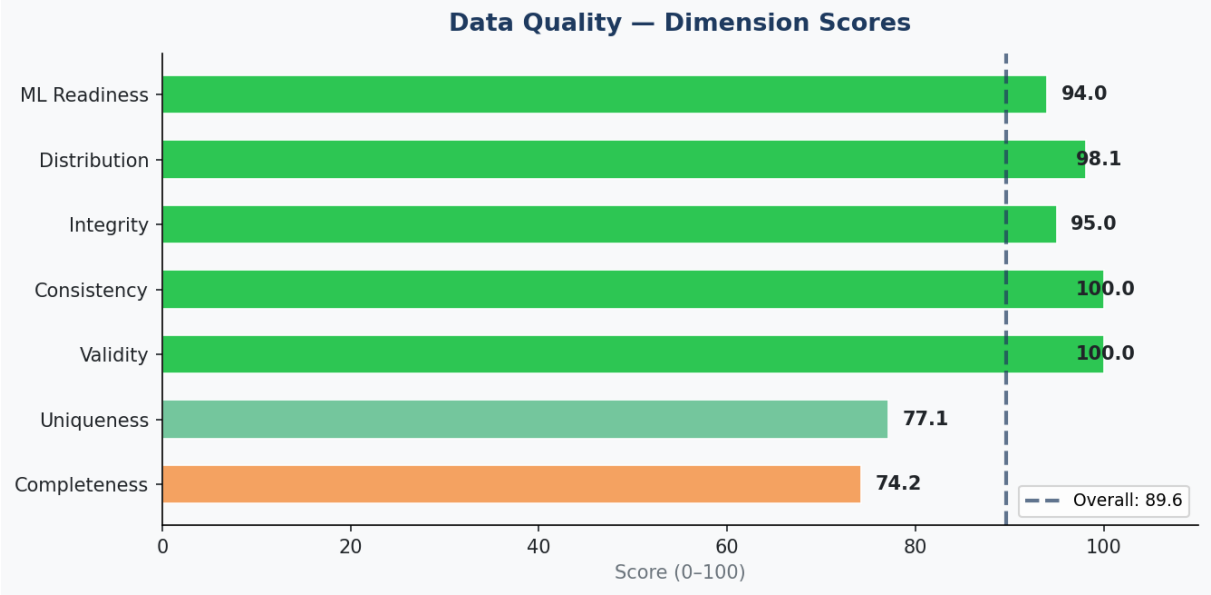
This report presents a comprehensive data quality analysis of the dataset **dataDoctor_big_dataset.csv**, evaluated across 7 quality dimensions using the dataDoctor Quality Scoring Engine. The dataset contains **10,300 rows** and **8 columns**.

The overall quality score is **89.6/100** (Good ■). A total of **4 issues** were identified across all dimensions, alongside **18 positive indicators**.

■ **PII ALERT:** Personally identifiable information was detected in 1 column(s). Review the Integrity section for details.

■ Dataset quality is good. Minor fixes recommended.

Dimension Scores Overview



2. Dimension Breakdown

Dimension	Score	Weight	Status	Issues
Completeness	74.2/100	22%	■ Good	1
Uniqueness	77.1/100	15%	■ Good	2
Validity	100.0/100	18%	■ Excellent	0
Consistency	100.0/100	15%	■ Excellent	0
Integrity	95.0/100	10%	■ Excellent	1
Distribution	98.1/100	10%	■ Excellent	0
ML Readiness	94.0/100	10%	■ Excellent	0

2.1 Completeness — 74.2/100

No column has >50% missing values ■

■ 4 column(s) missing 10-50%: quantity, region, sales_rep, total

2.2 Uniqueness — 77.1/100

No constant (zero-variance) columns ■

■ 300 duplicate rows (2.9% of dataset)

■ ID column(s) with duplicates: id

2.3 Validity — 100.0/100

All named numeric columns within expected ranges ■

No impossible negative values found ■

String column formatting appears consistent ■

2.4 Consistency — 100.0/100

Date columns appear logically ordered ■

Min/Max column pairs are consistent ■

No dtype mismatches detected ■

2.5 Integrity — 95.0/100

All column names are clean and well-formatted ■

Index integrity is clean ■

■ ■■ PII detected in 1 column(s): total

2.6 Distribution — 98.1/100

All numeric distributions have acceptable skewness ■

Outlier rates within acceptable bounds ■

Categorical columns have good value diversity ■

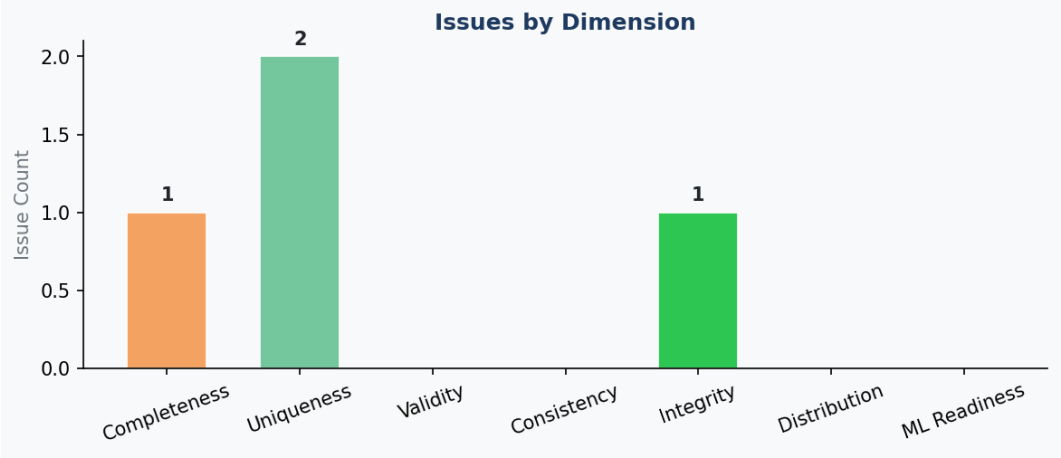
2.7 ML Readiness — 94.0/100

10,300 rows — sufficient for ML ■

4 numeric features available ■

Numeric features are well-populated ■

3. Issues & Recommendations



Prioritized Recommendations:

#	Recommendation
1	■ Dataset quality is good. Minor fixes recommended.
2	■ Impute moderate-missing columns using mean/median/KNN: quantity, region, sales_rep
3	■ Remove 300 duplicate rows → df.drop_duplicates()

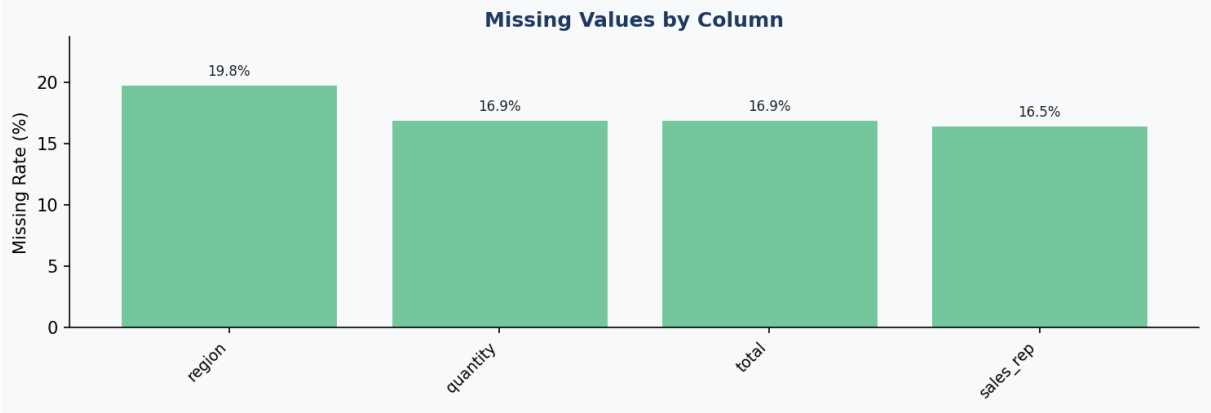
4. Dataset Statistics

Metric	Value
Total Rows	10300
Total Columns	8
Numeric Columns	4
Categorical Columns	4
Datetime Columns	0
Null Cells	7217
Overall Null Rate	8.76%
Duplicate Rows	300
Memory Usage (MB)	0.882

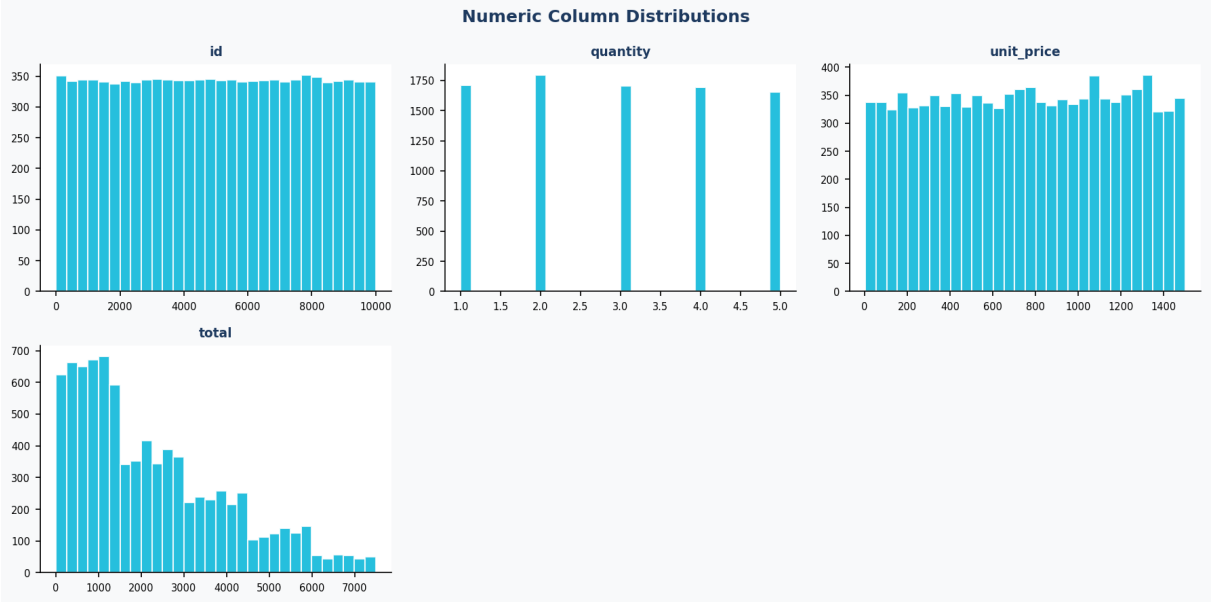
Column Summary

Column	Dtype	Null %	Unique	Most Common
id	int64	0.0%	10000	30
product	str	0.0%	8	Desk
category	str	0.0%	3	Furniture
quantity	float64	16.9%	5	2.0
unit_price	float64	0.0%	9671	220.76
region	str	19.8%	4	East
sales_rep	str	16.5%	5	Carol
total	float64	16.9%	8210	2796.88

5. Missing Values Analysis



6. Numeric Distributions



■■ 7. PII Risk Alert

The following columns may contain Personally Identifiable Information (PII). Review and apply appropriate anonymization, masking, or encryption before sharing or deploying this dataset.

Column	PII Signals Detected
total	pattern:phone, pattern:credit_card